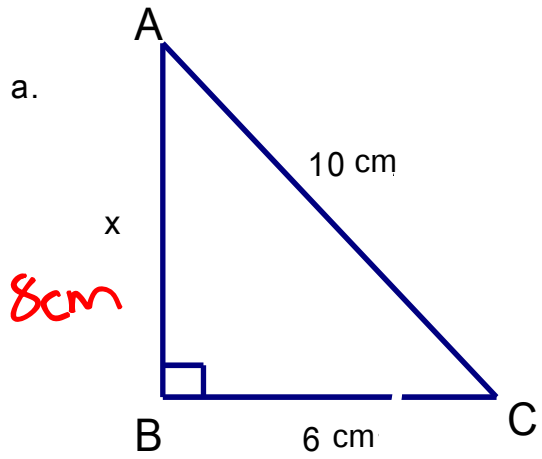
Exercise 1

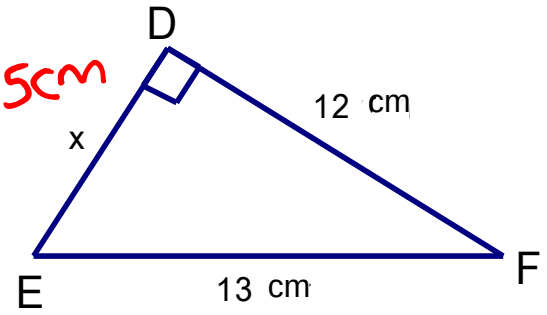
1. These diagrams are NOT drawn to scale. Where necessary give your final answers to 2 decimal digits. Calculate the lengths of the unknown sides. All measurements are in cm.

a.



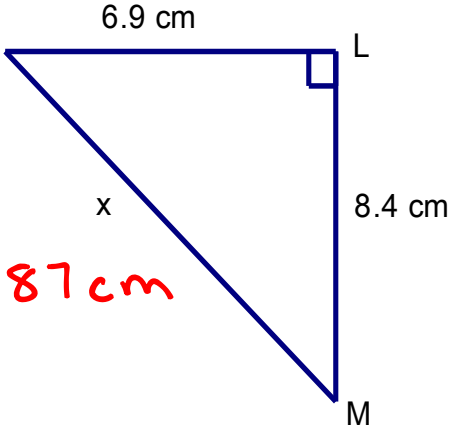
$x = 8cm$

b.



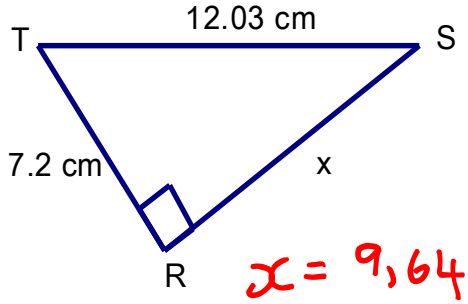
$x = 5cm$

c.



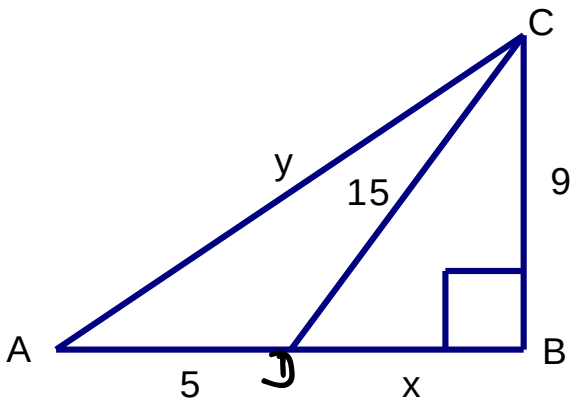
$x = 10,87\ cm$

d.



$x = 9,64$

2. Find the values of *x and y in the following. Show all your working and note that the diagram is not drawn to scale.*



In ΔCBD

$x^2 + 9^2 = 15^2$ Pythagoras

$x^2 = 144$

$x = 12$

In ΔABC

$y^2 = 9^2 + (5+12)^2$

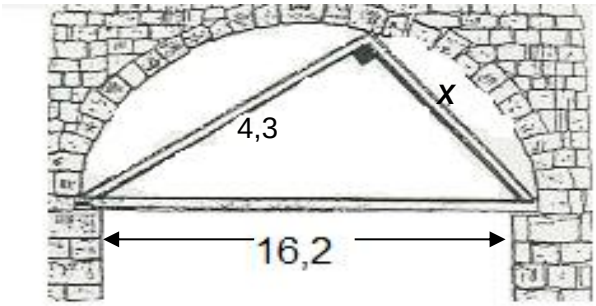
$= 9^2 + 17^2$

$\therefore y = 19,24\ cm$

Exercise 2 - Pythagoras in Action

1. Help the engineers find the missing information, leave your answer to TWO decimal digit.

a.

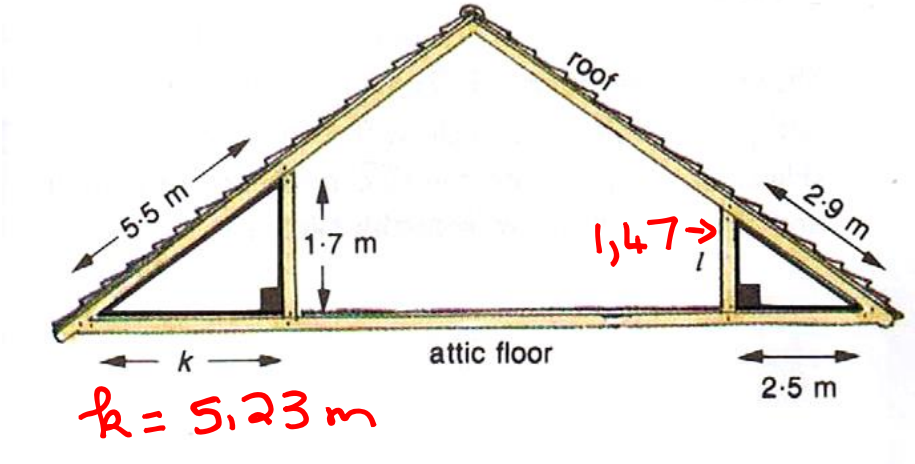


$x^2 + 4{,}3^2 = 16{,}2^2$

$x = 15{,}62\,m$

Pythagoras

b.



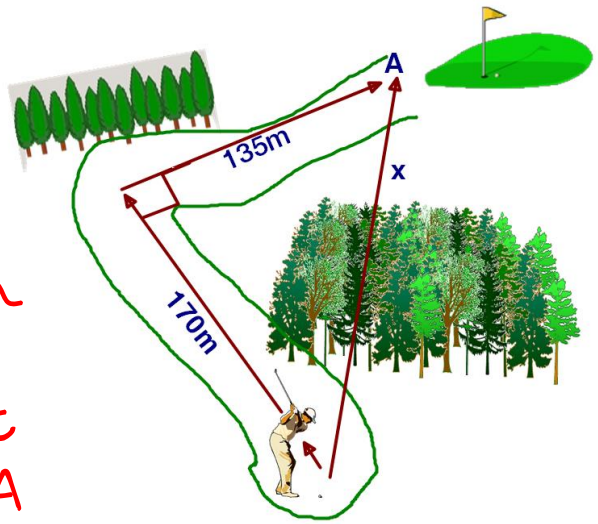
$k = 5{,}23\,m$

2. A golfer needs to know if she would be able to reach point A from the tee box. If she uses her driver she can hit the ball a distance of 210 m.
Will she reach point A?

Pythagoras

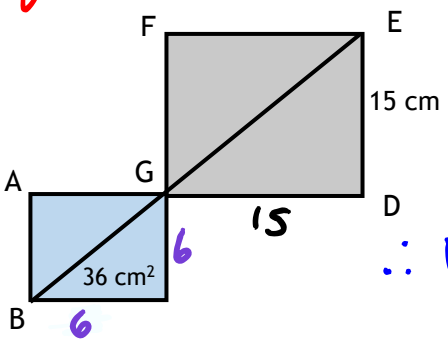
$x^2 = 135^2 + 170^2$

$x = 217{,}08\,m$



No, she will not reach point A
There is a difference of 7,08 m.

3. In the figure square ABCG has an area of 36cm², and square FEDG has sides with length 15 cm. Find the distance BE.



$BC = GC$ (sides of square)
$= 6$ cm

$BG^2 = 6^2 + 6^2$ (Pythag)

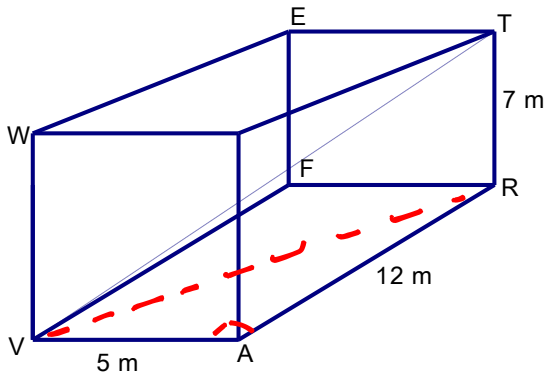
$BG \approx 8,49$ cm

$GF^2 = 15^2 + 15^2$
$= 450$

$\therefore GE = 21,21$ cm

$\therefore BE = BG + GE$
$= 8,49 + 21,21$
$= 29,70$ cm

4. The figure is a rectangular classroom, find the distance from the corner of the floor V, to the top of the roof T. Round off to two decimal digits.



In ΔVAR:

$VR^2 = 5^2 + 12^2$ Pythagoras
$= 169$

$\therefore VR = 13$m

In ΔVTR:

$VT^2 = TR^2 + VR^2$
$= 7^2 + 13^2$

$VT = 14,76$ m

Introduction to Trigonometry

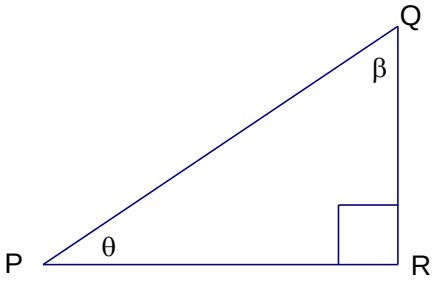
What is Trigonometry?
It is the study of the relationships between the sides and angles of triangles.

Trigonometry has its origins in astronomy. The beginning of trigonometry apparently was started by a man called Hipparchus in 150 BC. The Egyptians, the Babylonians and the early Greeks knew much about the ratios of angles and sides of triangles but it was Hipparchus who arranged the ratios in tables.

In trigonometry we normally make use of letters of the Greek alphabet to express the angles, example:

$\alpha = alpha; \quad \beta = beta; \quad \theta = theta; \; etc.$

Note that in ΔPQR: $\hat{P} = \theta \text{ and } \hat{Q} = \beta.$



Trigonometry on the Cartesian Plane:

Exercise 3:
Watch the video below and then answer the questions about the Unit circle:
https://www.youtube.com/watch?v=gQ44qpnXB2w *(9:56 min)*

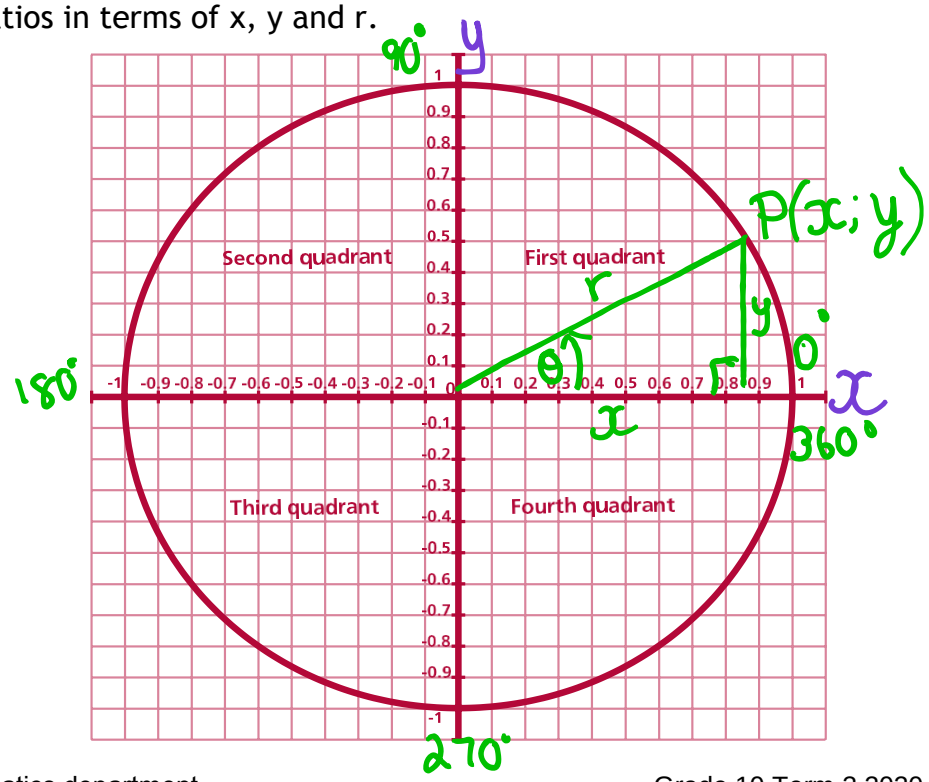
The UNIT CIRCLE:

1. In the unit circle, the radius is 1 unit. Label the $x \text{ and } y-axes.$
2. Fill in angles: $0^o; 90^o; 180^o; 270^o; 360^o$ and point $P(x;\ y)$
3. Join OP and drop a perpendicular to the x-axis.
4. Label angle θ and the sides $x, y \text{ and } r$ of the right angled triangle that is formed.
5. Write the 3 Trig ratios in terms of x, y and r.

$\sin\theta = \frac{y}{r}$

$\cos\theta = \frac{x}{r}$

$\tan\theta = \frac{y}{x}$



Trigonometry in the Right angled Triangle:

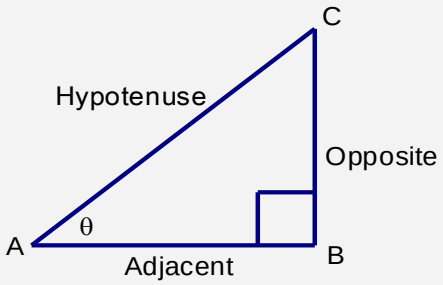


Figure 1

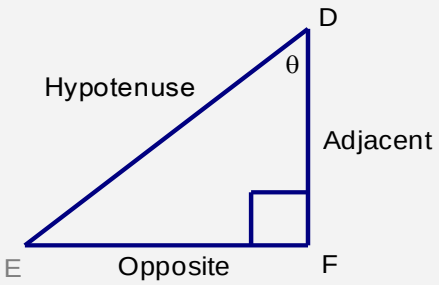


Figure 2

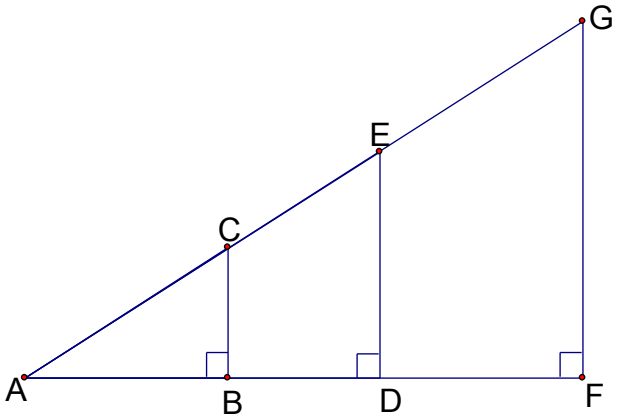
Let us revisit a right angle triangle.

In figure 1, side AB = adjacent because it is next to the angle θ and BC is opposite (across) the angle θ.

In figure 2, side DF = adjacent because it is next to the angle θ and EF is opposite (across) the angle θ.

Exercise 4

You will need a ruler for the following exercise:
Use the sketch and complete the following:



1 a. i) $\frac{CB}{AB} = \frac{.....}{.....}$ (ii) $\frac{ED}{AD} = \frac{.....}{.....}$ (iii) $= \frac{GF}{AF} = \frac{.....}{.....}$

b. Find the values of (i) ; (ii) and (iii) as a decimal, round off to 2 decimals.

c. What do you notice about these ratios? They are the same

These ratios will always be equal because of the **similar triangles**. We have a special name for this ratio; we refer to it as the **tangent** but abbreviate it:

$$\tan \hat{A} = \frac{\text{length of the side opposite } \hat{A}}{\text{length of the side adjacent to } \hat{A}}$$

2 a. i) $\frac{CB}{AC} = \frac{.....}{.....}$ (ii) $\frac{ED}{AE} = \frac{.....}{.....}$; (iii) $\frac{FG}{AG} = \frac{.....}{.....}$

b. Now, find the values of (i) ; (ii) and (iii) as a decimal round off to 2 decimals.

c. What do you notice about these ratios? They are the same

Likewise these ratios will always be equal. We also have a special name for this ratio; we refer to it as the **sine** but abbreviate it:

$$\sin \hat{A} = \frac{\text{length of the side opposite } \hat{A}}{\text{length of the hypotenuse}}$$

3 a. i) $\frac{AB}{AC} = \frac{.....}{.....}$ (ii) $\frac{AD}{AE} = \frac{.....}{.....}$; (iii) $\frac{AF}{AG} = \frac{.....}{.....}$

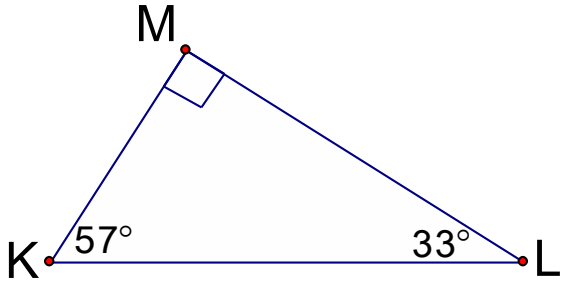
b. Now, find the values of (i) ; (ii) and (iii) as a decimal, round off to 2 decimals.

c. What do you notice about these ratios? They are the same

Similarly, they are equal in ratio. We refer to this ratio as the **cosine** of an angle but abbreviate it:

$$\cos \hat{A} = \frac{\text{length of the side adjacent to } \hat{A}}{\text{length of the hypotenuse}}$$

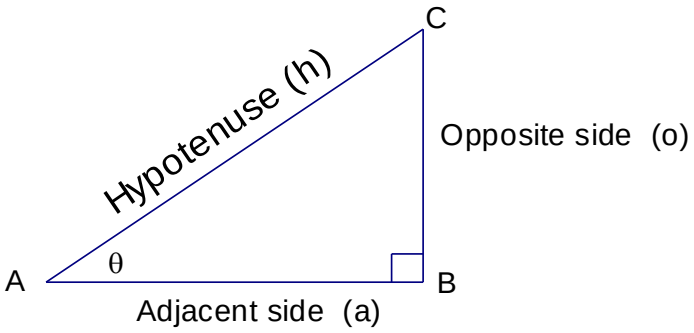
4. Use the following sketch ΔKLM, measure the sides KM, KL and ML, to the nearest millimetre.



5. Complete the table by using your measurements and the definitions of sin, cos and tan.

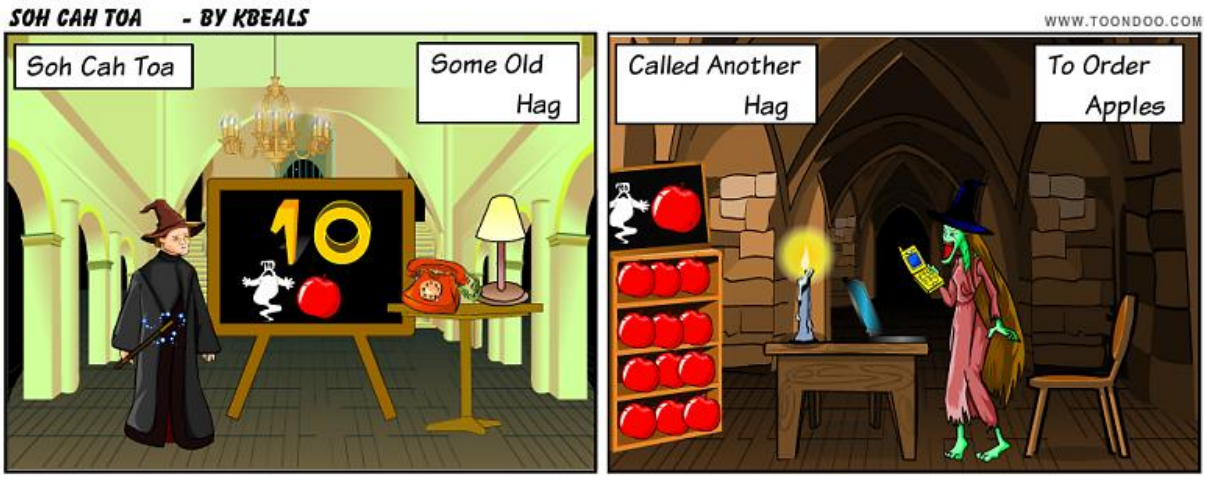
SIN	COS	TAN
$\sin 57° = \frac{\text{.....}}{\text{.....}}$ 0,839	$\cos 57° = \frac{\text{.....}}{\text{.....}}$ 0,545	$\tan 57° = \frac{\text{.....}}{\text{.....}}$ 1,54
$\sin 33° = \frac{\text{.....}}{\text{.....}}$ 0,545	$\cos 33° = \frac{\text{.....}}{\text{.....}}$ 0,839	$\tan 33° = \frac{\text{.....}}{\text{.....}}$ 0,649

To experience success in trigonometry you will have to find a way of remembering these ratios:



$$\sin\theta = \frac{Opposite}{hypotenuse} \qquad \cos\theta = \frac{adjacent}{hypotenuse} \qquad \tan\theta = \frac{Opposite}{adjacent}$$

SOH CAH TOA



Reciprocal Identities

$$\operatorname{cosec}\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

$$\sin\theta = \frac{1}{\operatorname{cosec}\theta} \qquad \cos\theta = \frac{1}{\sec\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

(These identities hold for any angle θ that does not lead to a 0 denominator.)

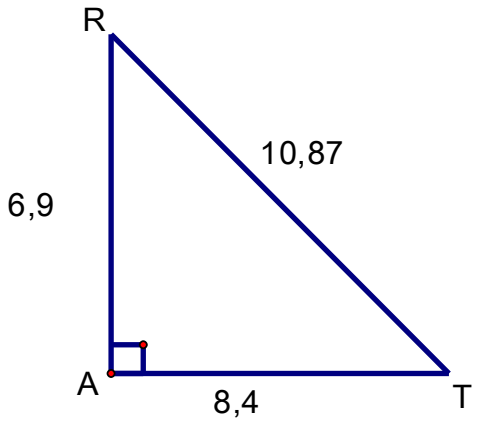
AP girls are required to work the inverse ratios a bit more……..Core Maths girls simply have to be aware of them.

Watch this video explaining SOH CAH TOA:

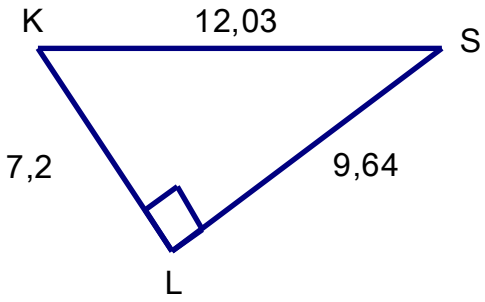
https://www.youtube.com/watch?v=hnrfdxySsQM

Exercise 5

Use the following triangles and complete the table, round off to ONE decimal digit:



b.



Figures not drawn to scale

sin R	cos R	tan R
0,8	0,6	1,2
sin T	cos T	tan T
0,6	0,8	0,8

sin K	cos K	tan K
0,8	0,6	1,3
sin S	cos S	tan S
0,6	0,8	0,7

Calculator work:

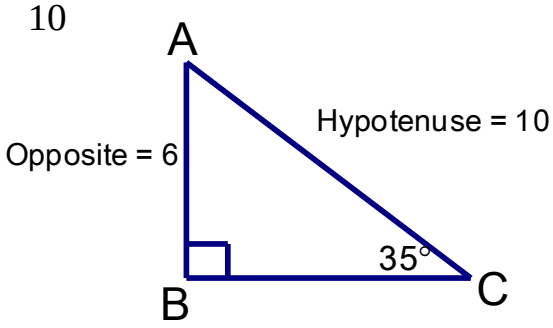
TYPE A: "Finding a RATIO when the angle is given"

Example 1:

<u>If we want to find a certain ratio</u>:

a. sin 35° = ?
sin 35° = 0.57357….. Rounded off to ONE decimal digit: sin 35° = 0,6

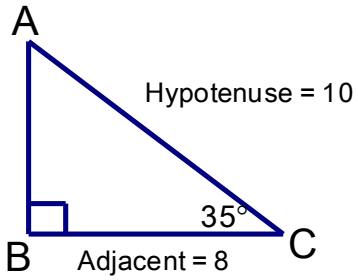
This means that sin 35° $\approx \frac{6}{10}$



b. cos 35° = ?

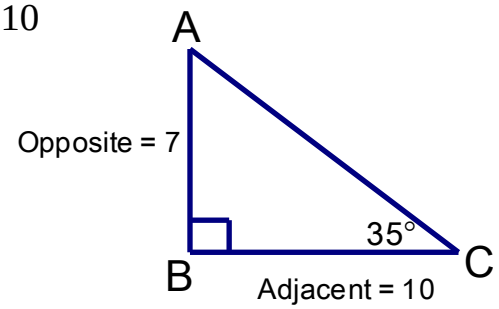
cos 35° = 0.81915…… Rounded off to ONE decimal digit: cos 35° = 0,8

This means that cos 35° $\approx \frac{8}{10}$



c. tan 35° = ?
tan 35° = 0.700207…… Rounded off to ONE decimal digit: tan 35° = 0,7

This means that tan 35° $\approx \frac{7}{10}$



Example 2:

Clearly see the difference between: (a) sin 2θ and (b) 2 sin θ.
If θ = 56° then:

a. sin 2θ = sin (2 X θ)
= sin (2 X 56°)
= sin (112°)
= 0.93 (Rounded off to 2 dec.)

b. 2 sin θ = 2 X sin θ
= 2 X sin 56°
= 2 X 0,82903.....
= 1,66 (Rounded off to 2 dec.)

NOTE:

$$\sin^2 30° \ \textit{means} \ (\sin 30°)^2$$

$$\therefore \ \sin^2 30° = (\sin 30°)^2$$

$$= (0,5)^2$$

$$= 0,25 \rightarrow$$

Exercise 7: "Finding Ratios when the angle is known"

Use your calculator and find the value of the following. Round off to <u>two</u> decimal digits.

1. a. tan 34° 0,67 b. sin 67° 0,92 c. cos 89° 0,02
d. sin 66,6° 0,92 e. cos14° 0,97 f. tan 3° 0,05
g. 3 sin 22,5° 1,15 h. 5 cos 76,9° = 1,13

2. a. 5,1 tan 68.93° 13,24 b. 89,72 cos 56,5° 49,52
c. $\sin\left(\frac{64,87°}{3}\right)$ 0,37 d. $\frac{\sin 64,87}{3}$ 0,30
e. tan 67° + tan10,5 2,54 f. tan(67°+10,5°) 4,51
g. Notice the difference between number 2e and 2f:
In e we have found the tan of two separate angles and then added.
In f we have added angles together first and then found the tan of the combined angles.

3. a. $\sqrt{\sin 30^\circ}$ 0,71 b. $\cos^2 45^\circ$ 0,5

c. $\tan 60^\circ + \cos 48^\circ$ 2,4 d. $4 \sin 123^\circ - \tan 93{,}7^\circ$ 18,82

e. $\sin^2 36^\circ + \cos^2 36^\circ$ 1 f. $\dfrac{\sin 167^\circ}{\tan 83{,}2^\circ}$ 0,03

g. $\tan^3 48{,}5^\circ$ 1,44 h. $2 \sin 117^\circ$ 1,78

4. If $A = 67{,}3^\circ$ and $B = 78{,}2^\circ$, find the value of the following:

a. $\cos A + \cos B$ 0,59 b. $\cos (A+B)$ -0,82 c. $\tan A + \tan B$ 7,18

d. $\tan (A+B)$ -0,69 e. $\sin A + \sin B$ 1,9 f. $\sin (A+B)$ 0,57

5. What do the examples in question 4 illustrate? $\cos A + \cos B \neq \cos(A+B)$ etc
Distributive law does not apply.

6. If $C = 53{,}8^\circ$ and $D = 78{,}2^\circ$, find the value of the following:

a. $\dfrac{\sin C}{\cos C}$ 1,37 b. $\tan C$ 1,37 c. $\dfrac{\sin D}{\cos D}$ 4,79 d. $\tan D$ 4,79

7. What do the examples in question 6 illustrate? Will it always be the case? yes
Use two examples of your own to verify your findings. $\dfrac{\sin\theta}{\cos\theta} = \tan\theta$

8. Find the value of the following:

a. $\sin 63^\circ$ 0,89 b. $\cos 27^\circ$ 0,89 c. $\sin 1^\circ$ 0,02

d. $\cos 89^\circ$ 0,02 e. $\sin 50^\circ$ 0,77 f. $\cos 40^\circ$ 0,77

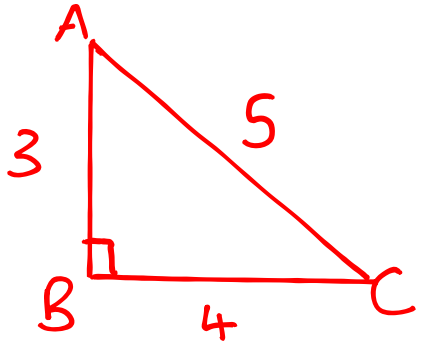
g. $\sin 44^\circ$ 0,69 h. $\cos 46^\circ$ 0,69

Co-ratios and complementary angles

$\sin A = \cos(90-A)$

$\cos A = \sin(90-A)$

9. What do the examples in question 8 illustrate? Will it always be the case?
Verify your answer by using any triangle ABC, with $\hat{B} = 90^\circ$, a = 4 cm, b = 5 cm and c = 3 cm.



$\sin \hat{A} = \frac{4}{5}$

$\cos \hat{C} = \frac{4}{5}$

$\sin \hat{A} = \cos \hat{C}$

and $\hat{A} + \hat{C} = 90^\circ$

TYPE B: " Finding an ANGLE when the ratio is known"

Example 3:

<u>If we have the ratio of the sides and we are looking for the angle</u>:

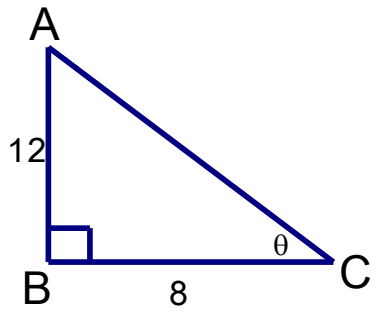
We have the choice of the sin , cos or tan function.
However, with the information given in the triangle we will have to use **tan.**

Why?_____

$$\tan\theta = \frac{opp}{adj} = \frac{12}{8}$$

$$\theta = \tan^{-1}\left(\frac{12}{8}\right)$$

$$\theta = 56,3° \quad (rounded\ \ off\ \ to\ \ ONE\ \ dec.)$$



$\tan^{-1}$ is obtained by pressing: **SHIFT** **tan**

Example 4:

Clearly see the difference between:

(a) $3\cos\theta = 1,246$ and (b) $\cos 3\theta = 0,896$

a. $3\cos\theta = 1,246$

$$\cos\theta = \frac{1,246}{3}$$

$\cos\theta = 0,4153333.....$ (*Don't round off yet!*)

$\theta = \cos^{-1} 0,4153333....$

$\theta = 65,46°$ (*Rounded off to 2 dec.*)

b. $\cos 3\theta = 0,895$

$3\theta = \cos^{-1} 0,895$

$3\theta = 26,49157.....$ (*Don't round off yet!*)

$$\theta = \frac{26,49157...}{3}$$

$\theta = 8,83°$ (*Rounded off to 2 dec.*)

Exercise 8:

1. Solve for θ in the following, round off to ONE decimal digit, θ ∈ [0°; 90°]

a.	tan θ = 3,2	b.	sin θ = 0,76	c.	cos θ = 0,6543
d.	sin θ = 0,123	e.	cos θ = 0,346	f.	tan θ = 64,78
g.	3 sin θ = 0,546	h.	2 tan θ = 18,54	i.	tan θ - 3 = 12,68
j.	sin θ + 8 = 9.567				

2. Solve for β in the following, round off to ONE decimal digit, β ∈ [0°; 90°]

a.	sin 4β = 0,876	b.	cos(β+10°) = 0,13	c.	3 tan β = 6,243
d.	cos 2β = 0,346	e.	2 tan β = 64,78	f.	3 cos β = 0,546
g.	tan 2β = 18,54	h.	sin β = 1,567	i.	cos β = 1,234

3. What do you notice about number 2(h) and 2(i). Any explanation?

The good questions we should ask are:

(i) Is it possible that sin θ >1?

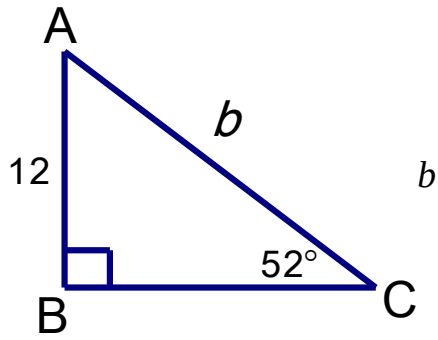
(ii) Is it possible that cos θ >1?

(iii) Why or why not? Think about the right angled triangle.

Example 5

To calculate the value of a side if you are given an angle and a side:

We have the choice of the sin , cos or tan function.
However, with the information given in the triangle we will have to use **sin. Why?**



$$\sin 52° = \frac{opp}{hyp}$$

$$\sin 52° = \frac{12}{b}$$

$$b \times \sin 52° = 12 \quad [Multiply\ both\ sides\ with\ b]$$

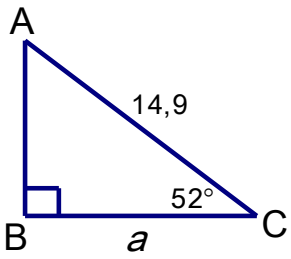
$$b = \frac{12}{\sin 52°}$$

$$\therefore b = 15,23 \quad (Rounded\ \ off\ \ to\ 2\ dec.)$$

Example 6

To calculate the value of a side, if you are given an angle and a side:

We have the choice of the sin , cos or tan function.
However, with the information given in the triangle we will have to use **cos. Why?**



$$\cos 52° = \frac{adj}{hyp}$$

$$\cos 52° = \frac{a}{14,9}$$

$$14,9 \times \cos 52° = a$$

$$a = 14,9 \cos 52°$$

$$\therefore a = 9,17 \quad (Rounded\ off\ to\ 2\ dec.)$$

Example 7

In this example, we will find the value of the variable that appears in the denominator

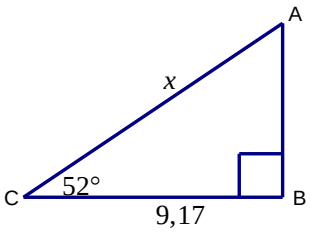
$$\cos 52° = \frac{adj}{hyp}$$

$$\cos 52° = \frac{9,17}{x}$$

$$x \times \cos 52° = 9,17$$

$$x = \frac{9,17}{\cos 52°}$$

$$\therefore x = 14,9 \quad (Rounded\ off\ to\ 1\ dec.)$$

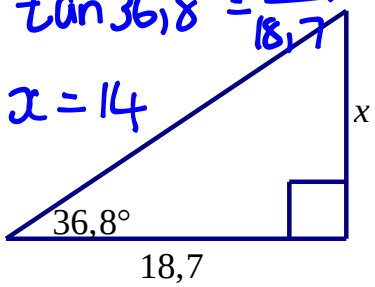


Exercise 9: Finding unknown sides of a triangle.

Find the value of *x* in the following right angle triangles. Round your answer off to ONE decimal digit:

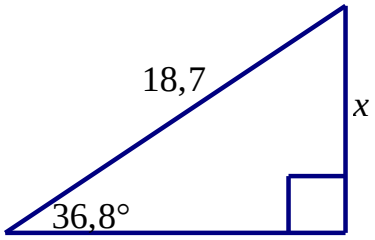
1. $\tan 36{,}8 = \frac{x}{18{,}7}$

$x = 14$



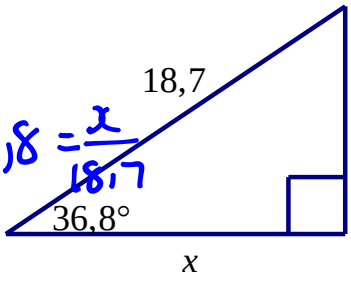
2. $\sin 36{,}8 = \frac{x}{18{,}7}$

$x = 11{,}2$



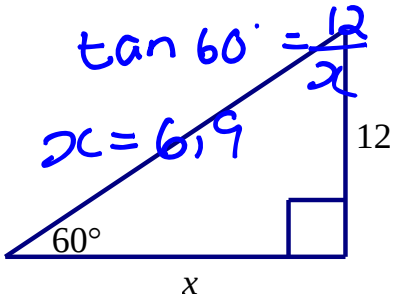
3. $\cos 36{,}8 = \frac{x}{18{,}7}$

$x = 15$



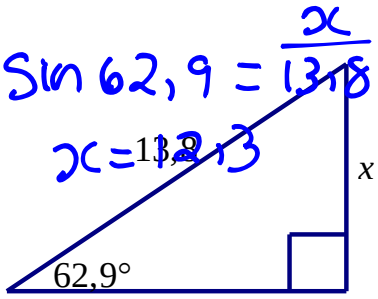
4. $\tan 60 = \frac{12}{x}$

$x = 6{,}9$



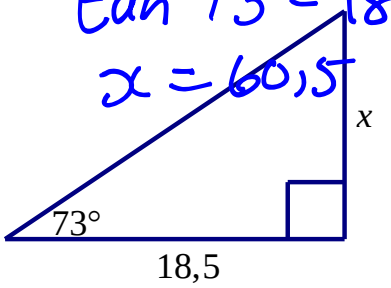
5. $\sin 62{,}9 = \frac{x}{13{,}8}$

$x = 12{,}3$



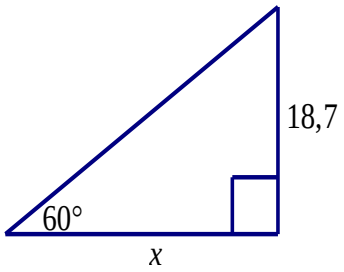
6. $\tan 73 = \frac{x}{18{,}5}$

$x = 60{,}5$



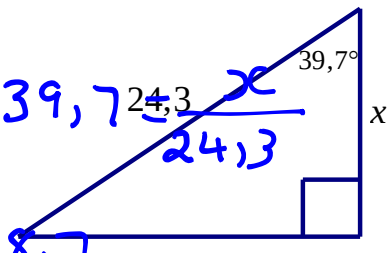
7. $\tan 60 = \frac{18{,}7}{x}$

$x = 10{,}8$



8. $\cos 39{,}7 = \frac{x}{24{,}3}$

$x = 18{,}7$



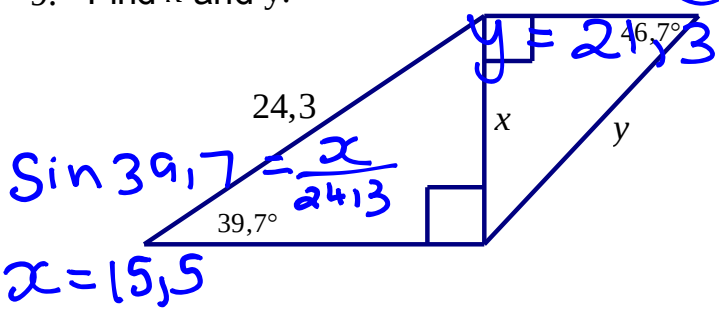
9. Find *x* and y.

$\sin 39{,}7 = \frac{x}{24{,}3}$

$x = 15{,}5$

$\sin 46{,}7 = \frac{15{,}5}{y}$

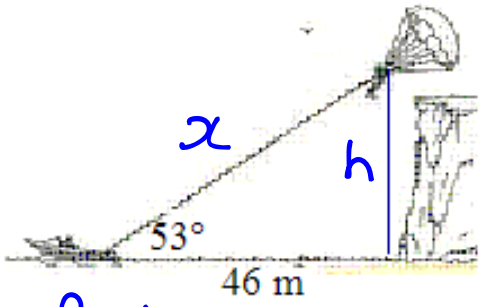
$y = 21{,}3$



Exercise 10 – Real life examples. Round answers off to <u>two</u> decimal digits.

1. Find: (a) the height of the Para-glider.

(b) the length of the rope.



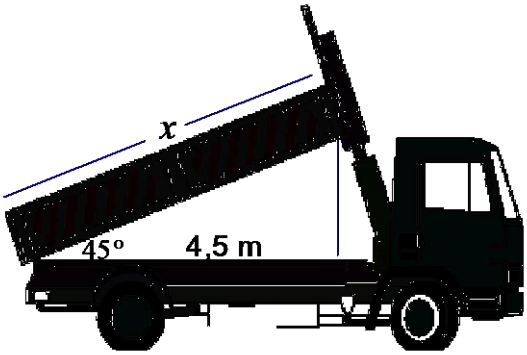
a) $\frac{h}{46} = \tan 53$

$h = 61{,}04 \text{ m}$

b) $x^2 = 61{,}04^2 + 46^2$ Pythagoras

$x = 76{,}44$

2. Sipho bought a new delivery truck. He knows the angle between the chassis load bed of the truck is 45°. He also knows that the length of the bed is 4,5m. Help Sipho find the length of the load bed.

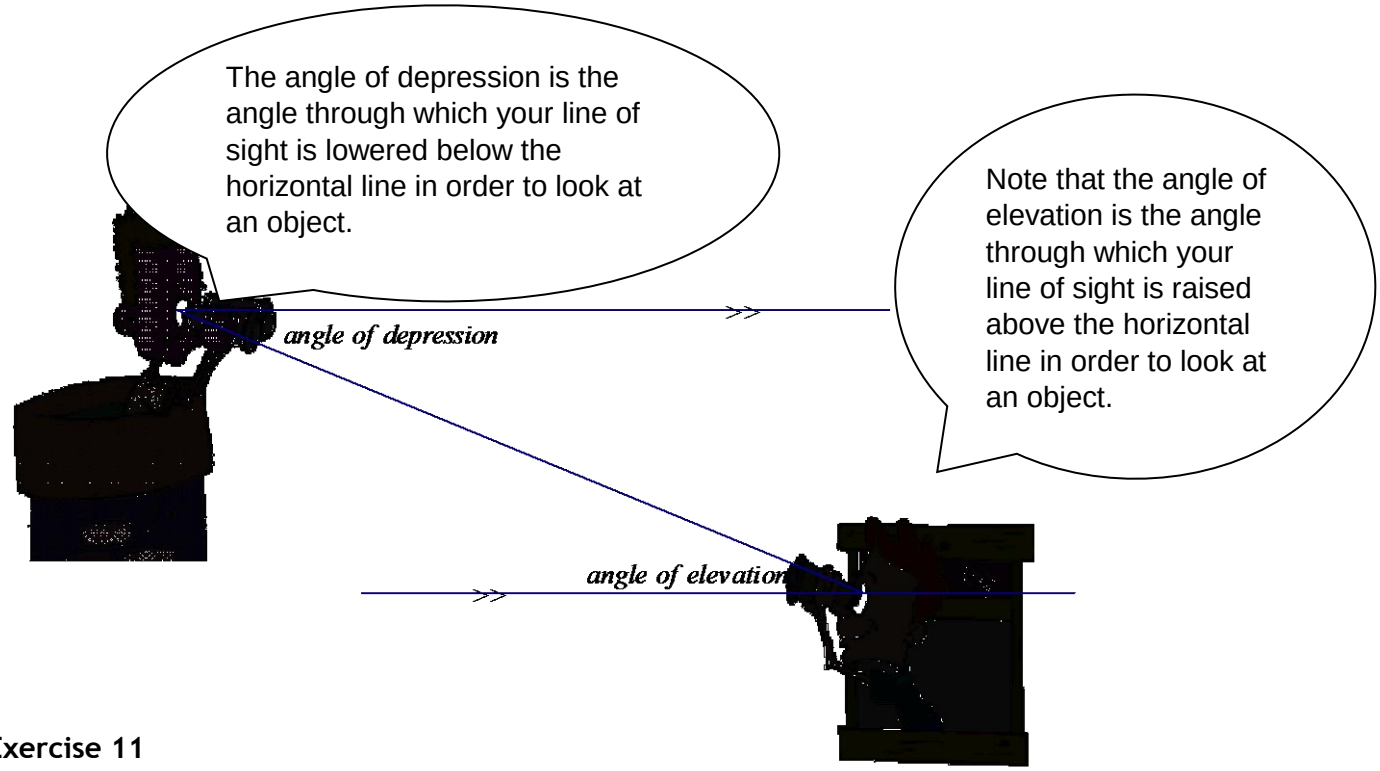


$\cos 45° = \frac{4{,}5}{x}$

$x = \frac{4{,}5}{\cos 45}$

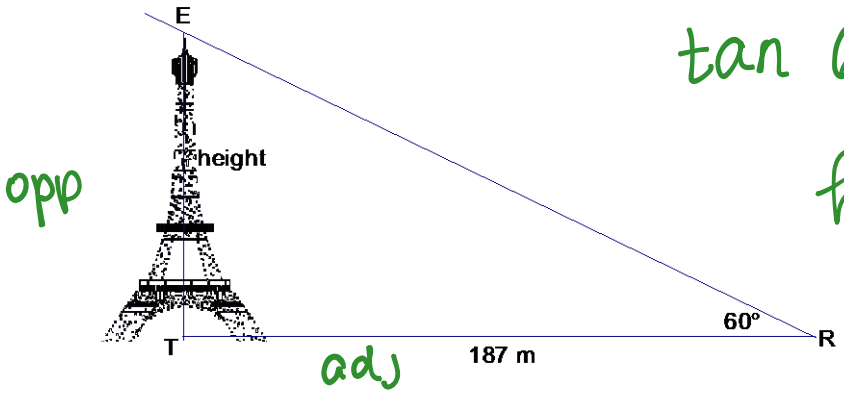
$x = 6{,}36 \text{m}$

Angle of elevation and depression



Exercise 11
Round answers off to <u>two</u> decimal digits.

1. The angle of elevation to the top of the Eiffel Tower, from point R, 187 m away from the tower, on the ground is 60°. How high is the Eiffel Tower?



$\tan 60° = \frac{h}{187}$

opp adj

$h = 323{,}89\ m$

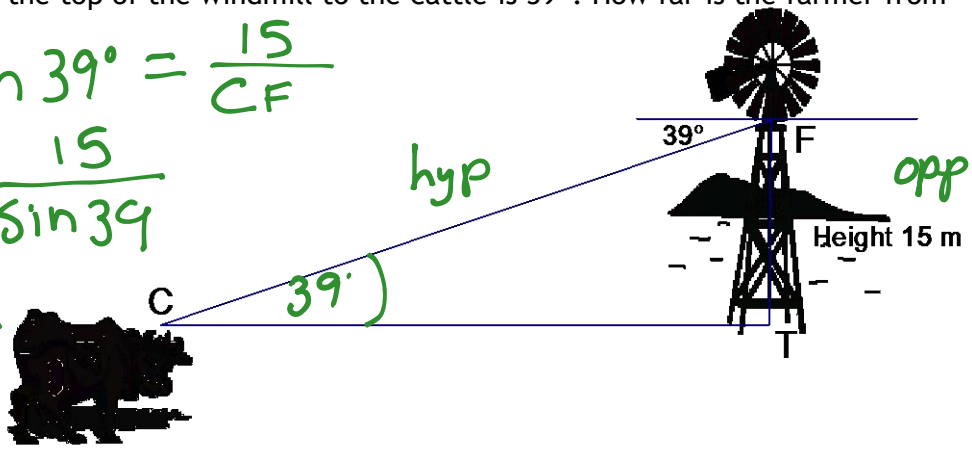
2. A farmer watches his cattle from the top of a windmill, 15 metres high. The angle of depression from the top of the windmill to the cattle is 39°. How far is the farmer from his cattle?

$\sin 39° = \frac{15}{CF}$

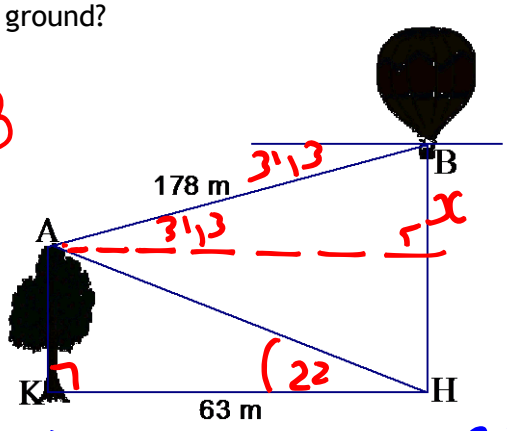
$CF = \frac{15}{\sin 39}$

$CF = 23{,}83\ m$

hyp opp



3. Peter is watching an apple tree from an hot-air balloon. He knows the distance between him and the apple tree is 178m. The angle of depression from the hot air balloon to the apple tree is 31,3°. The angle of elevation from the point H, on the ground, to the top of the apple tree is 22°.The hot air balloon is right above point H. How high is the air balloon from the ground?



$\frac{x}{178} = \sin 31{,}3$

$x = 92{,}47$

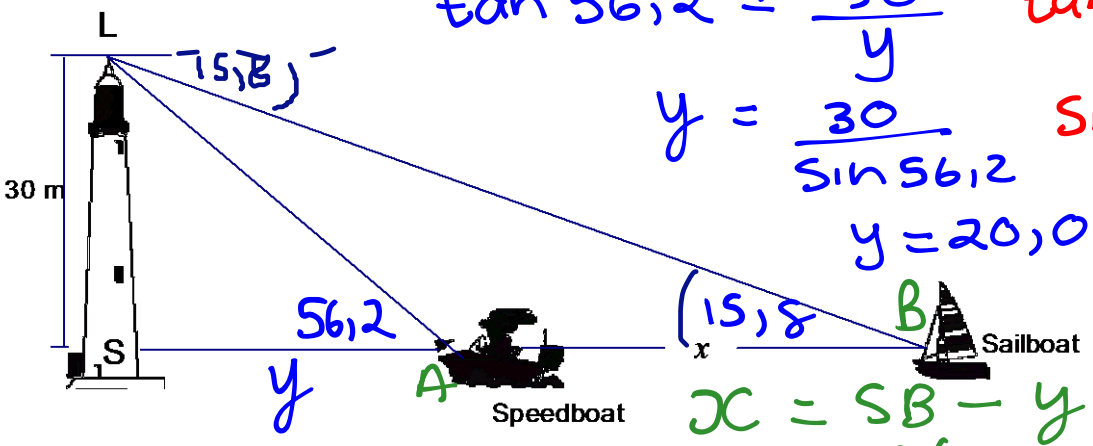
$\frac{AK}{63} = \tan 22$

$AK = 63 \tan 22$

$= 25{,}45\ m$

$BH = x + AK = 117{,}92\ m$

4. A lighthouse LS is 30 m high. The angle of depression from the lighthouse to the sailboat on the sea is 15,8°. The angle of elevation from the speed boat to the top of the light house is 56,2°. Find the distance between the speedboat and the sailboat.



$\tan 56{,}2 = \frac{30}{y}$

$y = \frac{30}{\sin 56{,}2}$

$y = 20{,}08$

$\tan 15{,}8 = \frac{30}{SB}$

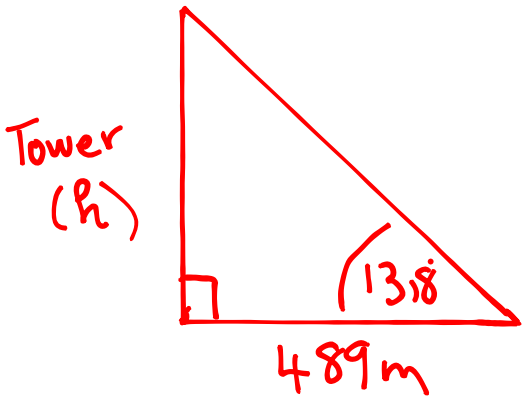
$SB = 106{,}02$

$x = SB - y$

$= 106{,}02 - 20{,}08$

$= 85{,}94$

5. The angle of elevation to the top of a tower is 13,8°. Tandeka is a distance of 489 m away from the foot of the tower. What is the height of the tower?
(Hint: draw your own sketch)



$\tan 13{,}8 = \frac{h}{489}$

$489 \tan 13{,}8 = h$

$120{,}11\ m = h$